CryptoGraph-Secure

Predicting crypto shock contagion with temporal graph neural networks

Viva and methodology guide: every phase, every technique, how we used it, and how to answer for it

How to use this guide. Start with the one-page cheat sheet. For each technique, read “what it is” and “how we used it”, then cover the page and say the green **30-second answer** aloud. The yellow **why this, not that** lines are what examiners probe. Practise with the self-test (section 17) by covering the answers.

Data: Binance hourly, 30 coins, 1 Jan 2020 – 27 Sep 2026. Code: branch feat/hourly-network-models (commit d327e0a). All numbers are from the project's notebooks 01–06, reports/ and runs/; test results are scored once per final model.

Confidential — possible paper / patent; do not share outside the project team and supervisor.

0. One-page cheat sheet

The pipeline in 6 steps

1. **Data:** 30 coins, hourly Binance prices 2020–2026 (1.61M coin-hours, 3,049 files checksum-verified), cleaned and audited.
2. **Shocks and events:** a shock = an hourly move > 2.29 × the coin's own recent volatility; every shock of coin i is an event towards every other coin j (1.46M events).
3. **Label:** does coin j also shock within k = 1, 3, 6, 12, 24 hours? (contagion)
4. **Network analysis:** correlation, spillover (Diebold–Yilmaz) and shock-transmission networks; centrality, communities, MST; independent-cascade baseline.
5. **Models:** TGN (main), DyGFormer, gradient boosting, LSTM, cascade, HAR-RV; tuned on validation, 5 seeds, each final configuration scored on test once (all such runs reported).
6. **Analysis:** significance tests, explanations (TempME, integrated gradients), shock persistence, robustness and T-GAE anomalies.

The models in one line each

Model	Type	What it does here
TGN	neural temporal graph (memory)	Main model: predicts contagion for each shock event
DyGFormer	neural temporal-graph transformer	Second graph model; ties the TGN
Gradient boosting	tabular ML, no graph	Answers "does the graph matter?"
LSTM	neural sequence model	Next-hour shock and volatility of one coin
Independent cascade	network diffusion (SNA)	Classic contagion baseline to beat
HAR-RV, logistic rules	econometrics	Volatility / shock baselines
Diebold–Yilmaz	VAR spillover network	Which coins transmit shocks (rankings)
TempME / integrated gradients	explainers	Why the TGN raises an alert
T-GAE	temporal graph autoencoder	Flags unusual market days

10 key numbers

#	Number	Meaning
1	0.205 vs 0.159 vs 0.105	Test AUPRC at 1 h: TGN vs cascade vs random
2	0.739	TGN test AUPRC at 24 h (cascade 0.672)
3	76%	of the TGN's riskiest 1% of alerts come true within 6 h (random 31%)
4	p ≈ 0.03	TGN beats gradient boosting on test at 1 h and 3 h
5	1.46 M	shock events at 1-hour resolution
6	z* = 2.29	shock threshold, chosen on training data only
7	64%	of hourly variance is one market factor
8	8–10 h	median time for a shocked coin's volatility to return to normal
9	65%	of an alert's attribution comes from the receiving coin's own state
10	13 of 15	riskiest alerts in LUNA, FTX and 10-Oct-2025 came true

3 honest limitations

- **Modest overall accuracy:** AUROC 0.64–0.69; most crypto shocks are news-driven and arrive market-wide within the same hour.
- **One market, one test period** (crypto, 2024–26); no social-media or order-book data.
- **Vulnerable to deliberate manipulation:** a white-box gradient attack of 0.1 std cuts AUPRC by about a third.

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1. Executive summary

Problem. When one cryptocurrency suffers a sudden shock, which other coins will follow, and how fast? We predict this contagion at the level of individual coin pairs and hours.

Data. Hourly prices of 30 major coins from the Binance public archive, 1 Jan 2020 – 27 Sep 2026, every file verified against its SHA-256 checksum; fixed-date splits: train to 30 Jun 2022, validation Jul 2022 – Dec 2023, test Jan 2024 – Sep 2026, with a 7-day gap (embargo) between them.

Headline results (test period, 5 seeds):

Model	1 h	3 h	6 h	12 h	24 h
TGN (main)	0.205	0.352	0.464	0.588	0.739
DyGFormer	0.208	0.351	0.463	0.587	0.736
TGN without memory	0.208	0.344	0.454	0.576	0.729
Gradient boosting	0.178	0.330	0.451	0.581	0.736
Ensemble (val-weighted)	0.199	0.330	0.465	0.596	0.748
Independent cascade	0.159	0.279	0.381	0.507	0.672
Random (positive rate)	0.105	0.206	0.310	0.447	0.627

AUPRC (area under the precision–recall curve); higher is better and random equals the positive rate.

Contributions. (1) A leakage-free hourly contagion benchmark (1.46M events) with significance tests that respect daily clustering. (2) Temporal graph models beat the classic cascade at every horizon and beat plain machine learning out of sample, especially after market-wide shocks. (3) Explanations show the signal lives in coins' current state, not in chains of past events. (4) Persistence, robustness and anomaly analyses with honest negative results.

2. Problem statement and research questions

Task	Unit	Question	Metric
T1 Contagion (main)	one shock event (i shocks at hour t), for every other coin j	Will j shock within k = 1, 3, 6, 12, 24 h?	AUPRC (vs positive rate), AUROC
T2 Shock risk	coin × hour	Will this coin shock in the next hour?	AUPRC, AUROC
T3 Volatility	coin × hour	Realised volatility over the next 1 / 6 / 24 h	MSE vs persistence, HAR-RV

Why crypto, and why hourly

What it is. Crypto trades 24/7 on one dominant exchange with public, checksummed tick archives, and has frequent crises (LUNA, FTX, SVB, the 10-Oct-2025 liquidation cascade).

How we used it. We moved from daily to hourly data after the first review: notebook 01 shows a shock makes another coin's shock **2.3–2.7× more likely in the next hour**, fading by 24 h.

> **Why this, not that:** Daily bars hide which coin moved first and merge several shocks into one; the 2-hour robustness check is weaker than 1 h (TGN 0.168 vs cascade 0.148), confirming 1 h as the right resolution.

> **Say it (30 s):** Crypto gives us a 24/7 market with clean public data and many real crises. Contagion there happens within hours: a shock makes another coin's shock about 2.5 times more likely in the very next hour. Daily data would hide that, so we built everything on hourly bars and kept 2-hour bars only as a robustness check.

3. Project workflow and phases

Phase	What we did	Output
A Housekeeping	gitignore, verified events catalogue, archived and documented the first notebooks' issues, proprietary licence	clean private repo
B Restructure	layered packages data / network / models / training / explain / robustness; removed placeholder code; output hashes unchanged	tested code base
E1 Universe	top coins by market cap → 30 model coins (≥ 2 years hourly, ≥ 90% test coverage); exclusions recorded	universe audit
E2 Hourly retrieval	Binance archive, SHA-256 checks, ticker changes, delistings, gaps, USDT peg, cross-check with daily data	1.61M coin-hours
E3 Preprocessing	features, seasonality, train-only scaling, shocks, contagion labels, splits, event stream	panel.pt, tgn_events.pt
E4 Hourly EDA	coverage, seasonality, lead–lag + Granger, Epps effect, contagion lift	notebook 01 Part II
E5 Network analysis	networks, centrality, communities, MST, crises, cascade baseline, network features	notebook 02
C Compute	device handling, seeding, run tracking with git commit, data packaging, Colab runner	runs/, results.csv
E6 Model plan	model table, protocol, feasibility of DyGLib / TempME	docs/model_plan.md
Training	baselines, TGN (v1 → v2 → v3), LSTM, DyGFormer, boosting, ensemble, refit, futures test	notebook 03
Explanations	TempME port, fidelity, integrated gradients, case studies	notebook 04
Persistence	fresh shocks by type, hours to normal, model performance by type	notebook 05
Robustness	T-GAE anomalies, noise, missing events, FGSM attacks	notebook 06

Engineering rules throughout: every run writes runs/<run_id>/ with its config, environment and git commit; final runs come from clean commits; each final configuration scores the test period once and every such run is reported (section 9); 127 automated tests and CI on every push; data and checkpoints never go into git.

Run tracking and reproducibility

What it is. Experiment tracking: each training run records exactly which code, config and seed produced each number.

How we used it. runs/<YYYYmmdd-HHMMSS>_<experiment>_<seed>_<commit>/ holds config.resolved.yaml, env.json (library versions, git commit, "dirty" flag), log, metrics.json, checkpoint and predictions; one row per run in runs/results.csv.

> **Why this, not that:** Notebook cells alone cannot prove which code made a figure; the commit hash can. It also caught a silent scripting bug (a PowerShell variable clash) that had broken one seed sweep.

> **Say it (30 s):** Every number in the project comes from a tracked run that records its git commit, config and seed. Final models only run from clean commits, and each final configuration scores the test period once. That is how we could catch and fix a silent sweep bug, and how any result can be reproduced.

4. Data: sources, universe and pitfalls

Sources

What it is. Binance public data archive (data.binance.vision): hourly spot candles, each monthly file with a SHA-256 checksum. Coin Metrics (daily USDT price for the peg check), CoinGecko snapshot (market-cap ranking for the universe).

How we used it. 3,049 files downloaded and verified, 0 failures; 2-hour bars resampled from 1-hour; daily data (Yahoo / Coin Metrics) only as a cross-check and context.

> **Why this, not that:** One exchange with verifiable files beats aggregated price feeds whose methods we cannot inspect; Yahoo rounds sub-cent coins (SHIB, PEPE) to 6 decimals, so it is unusable for them.

The 30-coin universe

What it is. Rule: a Binance USDT market, at least 2 years of hourly data and at least 90% coverage of the test period; stablecoins, wrapped/staked and pegged tokens excluded.

How we used it. 30 model coins across sectors (payments, smart-contract platforms, DeFi, meme, layer 2, exchange, privacy, AI); LUNC and FTT kept as event coins for case studies; XMR excluded (delisted Feb 2024); every exclusion recorded with its reason.

> **Why this, not that:** A coin without test-period data cannot be evaluated; mixing in stablecoins would add fake 'calm' nodes.

Pitfalls and how they were handled

Pitfall	Handling
LUNAUSDT is two different assets (old Terra LUNA, new LUNA)	Never spliced: old LUNA ends 13 May 2022; new-LUNA rows dropped; LUNC resumes Sep 2022; gap masked
MATIC → POL rename	79-hour gap masked; 1:1 migration, 0.0% price change across the switch
Delistings (XMR, FTT suspension)	Excluded / masked, never filled
Exchange maintenance gaps	≤ 2 h forward-filled and flagged; longer gaps masked (inactive)
Prices in USDT, not USD	Daily peg check from Coin Metrics (flags COVID Mar 2020, SVB Mar 2023); ends 23 May 2026 where the archive ends, stated as a limitation
Timestamps in ms, then µs from 2025	Normalised to UTC hours
Extreme hourly moves (92 with |log return| > 0.25)	87 confirmed by daily moves or catalogued events, 5 flagged, 0 bad ticks

Cross-check: the 23:00 bar of each day vs the daily close differs by about **0.1% (median)** for every coin; larger differences only in thin early markets (2020–21).

> **Say it (30 s):** Our data is hourly Binance prices for 30 coins from 2020 to 2026. Every file is checked against its SHA-256 checksum. We handled the traps explicitly: LUNA's ticker was reused for a different coin, so we never join them; MATIC became POL; delisted coins are masked, not filled. The hourly data matches independent daily data to about 0.1%.

5. Data preprocessing and analysis (notebook 01)

Notebook 01 has two parts: Part I on the daily data (the first phase) and Part II on the hourly data the models use. Each analysis led to a concrete modelling decision.

Quality audit and active masking

What it is. Checks for impossible prices, duplicates, gaps and outliers; a mask marking which coin is trading at each hour.

How we used it. Daily: 90,643 coin-days, 28 bad prices corrected. Hourly: no zero/negative prices or broken bars; 280 hours forward-filled, 10,779 masked. Coins listed later are simply inactive before listing (staggered launches), never back-filled.

> **Why this, not that:** Filling missing hours would invent returns and fake shocks; masking keeps them out of every loss and metric.

Log returns and stationarity (Augmented Dickey–Fuller)

What it is. ADF tests whether a series has a unit root (non-stationary, e.g. a random walk). Log return = ln(P_t / P_{t-1}).

How we used it. Prices fail the test (non-stationary); log returns pass, so all models work on returns, volatility and other stationary transforms, never on price levels.

> **Why this, not that:** Models trained on trending price levels learn the trend of the training years and fail later (spurious regression).

Fat tails (Jarque–Bera, kurtosis)

What it is. Jarque–Bera tests whether returns are normally distributed; kurtosis measures tail heaviness (3 for a normal distribution).

How we used it. Daily returns have kurtosis about **18**: extreme moves are far more common than a normal distribution allows. Consequences: shocks are defined relative to each coin's own volatility, inputs are robust-scaled and clipped, regression uses Huber loss.

> **Why this, not that:** Mean-and-variance thinking (and plain MSE) would be dominated by a few crash hours.

Autocorrelation: direction vs volatility

What it is. The ACF measures how a series correlates with its own past.

How we used it. Returns have almost no autocorrelation (direction is not predictable), but absolute returns / volatility are strongly autocorrelated (volatility clustering). So we predict shocks and volatility, never price direction; direction is only a sanity check.

> **Why this, not that:** Predicting direction would be predicting the market — the data says there is nothing there.

Intraday seasonality

What it is. Volatility and volume follow the clock: some hours of the day and days of the week are systematically busier.

How we used it. Volatility ranges from **0.69× to 2.17×** its average by hour-of-week; an hour × weekday profile fitted on training data only is divided out before computing features and shocks.

> **Why this, not that:** Otherwise every US-open hour would look like a 'shock' and models would learn the clock, not risk.

Train-only robust scaling

What it is. Standardising features using statistics (median, interquartile range) computed on the training period only.

How we used it. Scalers and the seasonality profile are fitted on 2020–22 and applied unchanged to validation and test.

> **Why this, not that:** Fitting scalers on all data leaks future information (e.g. 2024 volatility levels) into training.

Splits, embargo and sample masks

What it is. Time-ordered train / validation / test periods; an embargo is a gap between them; sample masks mark which (hour, coin) pairs may be scored.

How we used it. Train to 30 Jun 2022, validation to 31 Dec 2023, test to 27 Sep 2026; 7-day embargo; a sample counts only if its whole label window lies inside its own split.

> **Why this, not that:** Random splits let the model see the hour after an example during training (leakage). Fixed dates mimic real use: train on the past, predict the future.

Lead–lag and Granger causality (with Benjamini–Hochberg)

What it is. Lead–lag: does BTC or ETH move first? Granger causality: do past values of one series improve the prediction of another? BH controls the false-discovery rate across many tests.

How we used it. Same-hour correlation 0.65–0.68, largest lagged correlation only about 0.035; Granger: a few significant pairs but at most 0.47% extra explained variance, and the pattern does not replicate on validation.

> **Why this, not that:** So we did **not** add lead–lag edges: coins move together within the hour, not one after another. This finding returns in notebooks 02 and 04.

Epps effect and the graph threshold

What it is. The Epps effect: correlations measured on short intervals are weaker than on long ones.

How we used it. Weak here (mean correlation 0.58 at 1 h vs 0.59–0.62 at 2 h – 1 day). The correlation threshold for graph edges is fitted on training data for 30% density: **0.75 at 1 h**, 0.77 at 2 h.

> **Why this, not that:** A fixed threshold such as 0.5 would connect 75% of all pairs at 1 h — a meaningless, almost complete graph.

Correlation: why a graph model

What it is. Average pairwise correlation and how the network of strong correlations changes over time.

How we used it. Coins share most of their movement (daily correlation 0.57) and the network changes every day, especially in crises.

> **Why this, not that:** Independent per-coin models ignore exactly the interactions contagion is about; a temporal graph model represents them directly.

> **Say it (30 s):** Preprocessing turned raw prices into safe model inputs: returns instead of prices because prices are non-stationary, robust scaling because returns are fat-tailed, seasonality removed because volatility follows the clock, and everything fitted on training data only with an embargo between periods. The analyses also told us what not to model: direction is unpredictable, and there is no usable lead–lag, because coins move together within the hour.

6. How events are quantified

Shock definition

What it is. A z-score: $z_t = r_t / \sigma_{t-1}$, the hourly (deseasonalised) return divided by the coin's trailing 7-day (168-hour) volatility measured up to the previous hour. A shock is $|z| > z^*$.

How we used it. $z^* =$ **2.29**, chosen on the training period so that about **4%** of training hours are shocks (3.3% over all data). Both crashes and pumps count; the sign is kept as the direction.

> **Why this, not that:** A fixed % threshold (say 5%) would flag every hour of a volatile small coin and almost never BTC; dividing by the coin's own past volatility makes shocks comparable across coins and over time, and using only past volatility avoids look-ahead.

Events and their features

What it is. A continuous-time event stream, the input format of temporal graph networks: (source, destination, time, message).

How we used it. When coin i shocks at hour t, one event $i \rightarrow j$ is created for every other active coin j: **1,463,693 events** (361k train, 301k validation, 802k test). Message (5 numbers): lag-1 correlation and same-hour correlation of i and j (trailing 7 days), i's return, i's volatility z, i's shock flag. The shock flag is always 1 (every event is a shock of its source by construction), so **4 of the 5 features carry information**; it is kept only for format compatibility. Node features (14 per coin): 6 market features at t + 8 network features (section 7).

> **Why this, not that:** Only top-k receivers per shock was rejected: it would bias the labels towards pre-selected pairs. Every pair is scored instead.

Contagion labels

What it is. Binary labels per event and horizon.

How we used it. label_k = 1 if j shocks in hours t+1 … t+k, $k \in \{1, 3, 6, 12, 24\}$; masked when j is inactive or the window leaves the split. Positive rates on test: 10.5% (1 h) to 62.7% (24 h).

> **Why this, not that:** Strictly later hours only — counting j shocking in the same hour would label simultaneous co-movement as contagion (an error in the project's first notebooks).

Shock types, catalogued events and stress days

What it is. Ways of grouping shocks and days.

How we used it. **Breadth** = share of other coins shocking in the same hour: market-wide ≥ 30%, partial 10–30%, idiosyncratic < 10%; **direction** crash / pump; **fresh** shocks = no shock of the same coin in the previous 24 h. **11 catalogued crises** with verified sources (COVID 2020, China ban 2021, LUNA, Celsius, FTX, SVB, ETF approval, yen carry-trade unwind, 10-Oct-2025 …). **Stress day** = some hour with ≥ 70% of coins shocking (21% of days).

> **Say it (30 s):** We quantify an event as a shock: an hourly move larger than 2.29 times the coin's own recent volatility, using only past data, with the threshold set on training data. Each shock of coin i becomes an event towards every other coin, 1.46 million events in total, each with four informative features such as the pair's correlation and the size of i's move. The label is whether coin j shocks in the following 1 to 24 hours. We also type shocks by how many coins moved together and by direction.

7. Network / social network analysis (notebook 02)

We apply social-network-analysis methods to the market: coins are the actors, co-movement and shock transmission are the ties. (We do not use social-media data.)

Correlation networks and the market mode

What it is. Nodes = coins, edges = strong return correlations over a trailing window; the market mode is the first principal component (the factor moving all coins together).

How we used it. Daily snapshots of the trailing-168-hour correlation graph (threshold fitted on train). One market factor explains **64%** of hourly variance.

> **Why this, not that:** This is why communities had to be found after removing the market mode — on raw correlations everything is one block.

Minimum spanning tree (MST)

What it is. The tree connecting all coins with the strongest correlations and no cycles (Mantegna; Onnela's normalised tree length tracks how tightly the market is linked).

How we used it. MST degree is a node feature; in all **9 crises** the tree shrinks (coins pull together) and correlation, density, spillover and shock lift rise.

Diebold–Yilmaz spillover network (generalized FEVD)

What it is. Fit a VAR on returns; the forecast-error variance decomposition says what share of coin j's future uncertainty comes from shocks to coin i. TO, FROM, NET and the total spillover index (TSI) summarise it.

How we used it. 30-day windows on the 30 coins; TSI **91.6%**; NET and spillover PageRank give transmitter rankings used to validate explanations.

> **Why this, not that:** **Generalized FEVD, not Cholesky:** Cholesky results depend on the arbitrary ordering of 30 coins; generalized FEVD is order-invariant. Hourly NET turned out equal to eigenvector centrality (ρ = 1.00) because there is no lead–lag, so we also built a time-ordered shock-transmission network.

Centrality: PageRank and eigenvector

What it is. Eigenvector centrality: a coin is central if it is linked to central coins. PageRank: the same idea on a directed graph with a damping factor.

How we used it. PageRank on the directed DY network and on the shock-transmission network (P(j shocks ≤ 1 h after i), 90-day windows); rescaled to mean 1 over active coins; used as node features and explanation benchmarks.

> **Why this, not that:** PageRank only on directed networks; on an undirected correlation graph it adds nothing beyond degree.

Communities (Louvain), modularity and sectors

What it is. Louvain groups nodes to maximise modularity (more links inside groups than chance); NMI compares two groupings (0 = unrelated, 1 = identical).

How we used it. Seeded Louvain on correlations with the market mode removed; communities partly match sectors (meme coins, BTC/ETH + DeFi, new platforms, old payment coins): **NMI 0.47, p = 0.003**. But contagion ignores sectors: within- and cross-sector 1-hour contagion are both **9.3%** (except meme coins).

Crises, the FTX case and the independent cascade

What it is. The independent cascade model (Kempe, Kleinberg & Tardos, 2003): each active node gets one chance to activate each neighbour j with probability p_ij — the classic SNA model of diffusion.

How we used it. p_ij fitted on training shocks; scores each event by the chance that j is activated within k hours. Test AUPRC at 1 h **0.160** vs random 0.105 — the bar every model must beat. FTX case: the market moved in the **same hour** as FTT's biggest falls, not in a chain.

> **Why this, not that:** Beating the base rate is easy; beating a fitted diffusion model shows the neural model learned more than 'how many coins shocked'.

Network features for the models

How we used it. 8 leakage-free node features per daily snapshot (strength, eigenvector, MST degree, community size, DY PageRank, DY NET, shock PageRank, shock out-lift), always from the latest snapshot at or before t; rankings kept for evaluation only, never as inputs.

> **Say it (30 s):** Yes, we do social network analysis, with coins as the actors. We built correlation, minimum-spanning-tree, Diebold–Yilmaz spillover and shock-transmission networks, computed PageRank and eigenvector centrality, found communities with Louvain and compared them with sectors, and studied how the network compresses in crises. The classic independent cascade diffusion model is our network baseline, and eight network measures are inputs to the neural models.

8. Neural temporal models and baselines

TGN — Temporal Graph Network (main model)

What it is. Rossi et al. (2020): a graph neural network for continuous-time event streams. Each node keeps a **memory** vector updated by a GRU whenever it takes part in an event; an attention layer reads each node's most recent events; a head turns the embeddings into predictions.

How we used it. PyTorch Geometric TGNMemory (memory 64, time encoding 16), attention over each coin's **10 most recent events**, head on [embedding of i, embedding of j, node features of i and j at t, event message] → 5 logits (one per horizon). **70,021 parameters.** Training processes events in time order; memory and neighbour lists are updated **only after** a batch is predicted (this is what prevents look-ahead). Batches of about 200 never split an hour, so all events of one hour are predicted with the same information (a consistency rule, not a leakage rule: every shock of hour t is known when the hour closes). Focal loss, Adam (lr 5×10^{-4}), up to 100 epochs, early stopping on validation AUPRC at 1 h (patience 10), 5 seeds.

> **Why this, not that:** **TGN, not only an LSTM:** an LSTM sees one coin's history and cannot represent who interacts with whom. **TGN, not a snapshot GNN:** contagion happens within hours, and continuous-time memory avoids choosing a snapshot length.

Result. Best overall model: test AUPRC 0.205 (1 h) to 0.739 (24 h); beats the cascade at every horizon (bootstrap intervals above zero, e.g. +0.046 [0.019, 0.074] at 1 h).

> **Say it (30 s):** Our main model is a Temporal Graph Network. Every coin has a memory that is updated each time it is involved in a shock event, and an attention layer looks at its ten most recent events. For a new shock it combines the memories and current state of both coins and predicts contagion at five horizons. We train it strictly in time order and update memory only after predicting, so it never sees the future.

Training choices: focal loss, early stopping, min_epochs, seeds

What it is. Focal loss down-weights easy examples so rare positives matter more; early stopping keeps the epoch with the best validation score.

How we used it. Focal loss ($\gamma = 2$, $\alpha = 0.25$). The first final run (30 epochs, patience 4) stopped while validation was still rising (0.138), so v2 used 100 epochs and patience 10 (0.153). min_epochs = 30 for models with a noisy early peak. Every final model: 5 seeds, mean ± std.

> **Why this, not that:** **Focal, not plain BCE:** with 10% positives at 1 h, plain BCE is dominated by easy negatives. Multiple seeds show whether a difference is real or seed noise (± 0.003–0.005). **Limitation:** early stopping uses the 1-hour score only, while memory helps most at 6–24 h; a multi-horizon criterion might pick other epochs.

DyGFormer

What it is. Yu et al. (NeurIPS 2023): a Transformer over each node's recent interactions, with neighbour co-occurrence encoding and patching; strong on temporal-graph benchmarks.

How we used it. Compact re-implementation (24,021 parameters, 16-event histories) on exactly the same events, splits and loss as the TGN.

> **Why this, not that:** The official library (DyGLib) uses random link-prediction negatives and quantile splits, which would break our leakage rules, so it was re-implemented. A second architecture shows the results do not depend on one model.

Result. Test 0.208 at 1 h, 0.736 at 24 h — tied with the TGN (no bootstrap difference at any horizon).

Gradient boosting ("does the graph matter?")

What it is. Histogram gradient boosting (scikit-learn, LightGBM-style): an ensemble of decision trees, the strongest standard model for tabular data.

How we used it. Exactly the TGN head's inputs without memory or attention, plus coin ids and market context; number of trees chosen on validation per horizon.

Result. Matched the TGN on validation (0.155 vs 0.153) but fell behind on test: **0.178 vs 0.205 at 1 h** ($p = 0.028$) and 0.330 vs 0.352 at 3 h; it memorised the training period (train AUPRC 0.64). After market-wide shocks the TGN wins clearly (0.245 vs 0.205); after idiosyncratic shocks they are level.

LSTM, HAR-RV and rule baselines (T2, T3)

What it is. LSTM: a recurrent network for sequences. HAR-RV (Corsi, 2009): tomorrow's volatility as a linear combination of recent daily, weekly and monthly volatility.

How we used it. LSTM on each coin's last 168 hours (5,665 parameters). T2 shock risk: test AUPRC **0.109** (AUROC 0.706), 2.8× random, above the best rule-based model (0.095). T3 volatility: **HAR-RV beats the LSTM** at every horizon (MSE 5.0×10^{-5} vs 5.5×10^{-5} at 1 h).

> **Why this, not that:** Reported honestly: for single-coin volatility the simple linear model is the right tool; deep learning earns its place on rare events and networks.

Other neural temporal components

- **TempME** (temporal motif explainer) — section 11.
- **T-GAE** (graph convolution + GRU autoencoder over daily graphs) — section 12.
- **Ablation:** the TGN without memory and attention matches it at 1 h (0.208) but is worse at 6–24 h (significant): memory captures slower, multi-hour contagion.

> **Say it (30 s):** Yes, neural temporal analysis is the core. The main model is a Temporal Graph Network with continuous-time memory; DyGFormer is a temporal transformer we use as a second graph model; an LSTM models single-coin sequences; a temporal graph autoencoder with a GRU detects unusual days; and TempME explains the TGN with temporal motifs. All are trained in strict time order, with five seeds, tuned on validation only.

9. Evaluation methodology

AUPRC against the positive rate (not accuracy)

What it is. AUPRC averages precision over all recall levels; for a random model it equals the share of positives.

How we used it. Reported per horizon next to the positive rate; AUROC as a secondary metric.

> **Why this, not that:** At 1 h, 90% of events are "no contagion": a model that always says no is 90% accurate and useless. AUROC is also optimistic with rare positives; AUPRC focuses on finding the positives.

Day-block bootstrap

What it is. Resampling with replacement to get confidence intervals; here whole days are resampled (971 test days, 1,000 times).

How we used it. Compares two models on exactly the same test events; reports the AUPRC difference, a 95% interval and a one-sided p-value. Written as a fast single-pass weighted AUPRC (checked against scikit-learn).

> **Why this, not that:** **Day blocks, not i.i.d. events:** one crash creates hundreds of correlated events; resampling single events would give falsely narrow intervals.

Fit diagnostic (over- or underfitting)

What it is. Lift = AUPRC / positive rate; ratio = train lift / validation lift.

How we used it. The cascade (cannot memorise) has ratio ≈ 1.1, which is the part due to the market changing; the TGN ≈ 1.5 at 1 h, so some short-horizon overfitting, controlled by early stopping; ≈ 1.1 at 24 h.

How the test period was used (and how to defend it)

What it is. Every configuration was fixed on validation before its single test run; but several configurations were scored on test over the project.

How we used it. Test-scored TGN versions: **v1** (30 epochs, patience 4), **v2** (100 epochs, patience 10; the main model), the 2-hour v2, the no-memory ablation (v1, v2, v3) and the refit. Others: LSTM finals, DyGFormer, boosting and its refit, the ensemble, T-GAE, stress tests. Every one is reported, including the worse ones.

> **Why this, not that:** Defence: (1) v1 → v2 was decided from the **validation** curves (still rising at epoch 30), not from test; the no-memory v2 → v3 change came from seeds stopping at epoch 1 on validation. (2) The main model was fixed before the refit and ensemble were tried, and we did not switch to anything that looked better on test (DyGFormer 0.208 and an equal-weight ensemble were not adopted). (3) Every version is reported, not just the best. (4) A rolling second test window is the clean fix and is first in the paper plan.

Test protocol and top-alert precision

How we used it. All tuning on validation; each final configuration scores the test period once (evaluate_test: true). For usefulness: precision of the riskiest 1% of alerts — 32% (1 h), 61% (3 h), **76% (6 h)**, 89% (24 h) vs 10.5 / 21 / 31 / 63% at random.

> **Why this, not that:** Looking at test results while tuning silently overfits the test set; one look keeps the test an honest estimate.

> **Say it (30 s):** We never use accuracy, because 90 percent of events are no-contagion. We report AUPRC against the positive rate, which is what random guessing scores. Significance comes from a bootstrap that resamples whole days, since one crash creates hundreds of correlated events. Everything is tuned on validation, each final configuration touches the test once, and every version that did is reported. In practice, three out of four of the model's top alerts come true within six hours.

10. Results (notebook 03)

Main T1 table: see the executive summary. Further results:

Question	Finding (test unless stated)
TGN vs cascade	Significant at every horizon; e.g. +0.084 [0.051, 0.116] at 6 h; no resample favoured the cascade
Does memory help? (ablation)	No at 1 h (0.208 vs 0.205); yes at 6–24 h (+0.011–0.013, p ≤ 0.025)
TGN vs DyGFormer	Tied at every horizon (all intervals include zero)
TGN vs gradient boosting	TGN better at 1 h and 3 h (p ≈ 0.03); boosting matched it on validation only
Ensemble (weights chosen on validation)	Better than the TGN at 24 h (+0.008, p = 0.001); worse at 1–3 h
Refit on train + validation	Helped boosting (0.178 → 0.185); hurt the TGN (0.205 → 0.188) — the train-only TGN stays the main model
Futures data (funding, premium)	Lowered validation AUPRC (TGN 0.153 → 0.131): leverage regime changed; not used
Tuning attempts (LR schedule, context, regularisation)	Within ± 0.005 = seed noise; simplest well-trained model kept
1 h vs 2 h	TGN beats the cascade at 2 h too (0.168 vs 0.148), but by less: 1 h is the right resolution
Seed stability	5 seeds agree within ± 0.003

Say it (30 s): On the 2024–26 test period the TGN reaches an AUPRC of 0.205 at one hour, against 0.159 for the cascade model and 0.105 for random, and it beats the cascade at every horizon with day-block significance tests. DyGFormer ties it. Gradient boosting on the same inputs matched it on validation but fell behind on test, so the graph model generalises better. Memory matters for contagion over six to 24 hours.

11. Explanations (notebook 04)

TempME

What it is. Chen & Ying (NeurIPS 2023): explains temporal graph models with **temporal motifs** (short time-ordered chains of events). A small network scores each motif; the scores re-weight the model's attention; trained with an information-bottleneck loss (keep the prediction, use as few motifs as possible, compared with a null model of motif frequencies).

How we used it. Ported to our TGN following the official code: motifs up to 3 steps from both coins, 16 motif categories, null model from a time-shuffled stream; importance multiplies the TGN's attention exactly. Captured inputs reproduce the model's predictions to 2×10^{-7}. Trained on 10,000 validation events, evaluated on test.

> **Why this, not that:** **Ported, not run as-is:** the official code only explains its own models on a link-prediction task. One deliberate change: the loss targets the model's probability, because its hard-label rule ($p > 0.5$) would make almost every contagion label 0.

Result. Characterization score with 3 events kept: TempME **0.157**, random 0.152, recency 0.192, best possible per-alert mask 0.248. Reason: the TGN's attention over recent events is **perfectly uniform** (entropy 1.000), and motif types match the time-shuffled null — shocks spread market-wide within the hour, not along chains.

Faithfulness metrics

What it is. Fidelity− = change in prediction when only the explanation is kept (lower = sufficient); Fidelity+ = change when it is removed (higher = necessary); characterization = their harmonic mean (Amara et al., 2022).

How we used it. Computed at 1–10 kept events, normalised by the effect of removing all recent events, against random, recency, the model's attention and an optimised mask.

Integrated gradients (feature level)

What it is. Sundararajan et al. (2017): attribution of a prediction to each input by integrating gradients from a baseline (here the training mean); the attributions add up to the change in the logit.

How we used it. On the TGN's prediction head for the 6-hour horizon; completeness checked to 0.009.

Result. **Receiving coin's state ≈ 65%** (volume surge vs its past week 25%, 24 h volatility 12%, latest return 9%), source coin ≈ 25%, shock event ≈ 10%, memory + recent events ≈ 7%. Same pattern in LUNA, FTX and 10-Oct-2025, where 13 of the 15 riskiest alerts came true.

> **Say it (30 s):** We ported TempME, the temporal-motif explainer, to our model and tested it on the real predictions. It was barely better than random, and we found why: the model weighs its recent events equally, and motif patterns look the same as in a shuffled market, because crypto shocks hit everything in the same hour. Integrated gradients then showed what drives an alert: mainly the receiving coin's own state, its volume surge and volatility. So the model flags coins that are already stretched when another coin crashes.

12. Persistence (notebook 05) and robustness (notebook 06)

Shock persistence by type

What it is. How long a shock's effect lasts, measured on 18,171 fresh shocks with day-block confidence intervals.

How we used it. Hours until the coin's mean |z| over 6 hours is back to its level of the previous day; re-shock within 24 h; followers within 6 h; continuation or reversal of the price move.

Result. Volatility back to normal after a median of **8–10 hours** for every type (87% within a day), the same in every period. 55–65% re-shock within 24 h. Market-wide shocks are followed by about **8** other coins shocking within 6 h vs about 5 after idiosyncratic ones. Direction mostly unpredictable (pumps tend to continue). **The TGN beats boosting after market-wide and partial shocks, not after idiosyncratic ones.**

T-GAE anomaly detection

What it is. A temporal graph autoencoder: graph convolution over each day's correlation graph, a GRU across days, decoders for the graph and each coin's daily activity; the anomaly score is the prediction error (how surprising a day is).

How we used it. Trained on 2020–22, 5 seeds; evaluated against independent stress days (≥ 70% of coins shocking in an hour).

> **Why this, not that:** Two designs were tried on validation (single-hour features with reconstruction; daily activity with prediction) and both are reported rather than tuned until something worked.

Result. Negative result: test AUROC 0.511 for stress days vs 0.893 for plain turbulence; adds nothing as a next-day warning (0.490 vs 0.600). It does rank the 10-Oct-2025 cascade as the most anomalous test day. Other top anomalies match no catalogued event and are reported as "unclassified" — no manipulation type is invented.

Stress tests

How we used it. Final TGN vs final boosting on test: Gaussian noise on inputs, missing events in the TGN's memory, white-box FGSM attack (moves inputs by ±ε in the direction that hurts most) transferred to boosting.

Result. Missing up to **50%** of events: TGN changes by < 0.5%. Noise σ = 0.5: TGN −5.7%, boosting −4.7%, TGN predictions steadier. FGSM ε = 0.1: TGN **−35%**; random noise of the same size costs nothing, so the risk is targeted manipulation. Defences (multi-source input checks, ensembles, adversarial training) are future work.

> **Say it (30 s):** Shocks fade within about ten hours but cluster, and market-wide shocks pull in more followers; that is also where the graph model beats plain machine learning. On robustness, the TGN ignores missing data and handles noise as well as boosting, but a deliberate gradient attack hurts it, so manipulation is the real risk. Our temporal graph autoencoder did not recognise stress days beyond simple turbulence, and we report that as a negative result.

13. Problems faced, honest negatives and limitations

Problem	How we solved it
First notebooks: look-ahead in shock z, co-movement labelled as contagion, random splits	Trailing-only shocks, strictly-later labels, fixed-date splits with embargo; documented and archived
LUNA ticker reuse, MATIC → POL, delistings, gaps	Never spliced, masked, recorded
1.46M events for a laptop CPU	Measured first, compared options, compact models, parallel seeds
TGN must never use future events	Memory and neighbour lists updated only after each batch is predicted; batches never split an hour (consistency)
One event feature carried no information (shock flag always 1)	Found in a code review; documented (4 informative features); no effect on results
Under-trained first final run	v2 budget (100 epochs, patience 10), min_epochs rule
Correlated events make significance optimistic	Day-block bootstrap
Boosting tied on validation	Test showed the graph model generalises better ($p \approx 0.03$)
Silent sweep bug, laptop sleep	Run registry checks; keep-awake; outputs verified, not only exit codes

Negative results we report: refit hurt the TGN; futures data hurt validation; TempME barely beats random; T-GAE has no stress signal; LSTM loses to HAR-RV for volatility.

Limitations: AUROC 0.64–0.69; one market and one test period, scored by several model versions (all reported); early stopping on the 1-hour score only; no social-media / order-book data; vulnerable to white-box attacks; USDT peg check ends May 2026.

Future work: (1) social-media sentiment as node features; (2) other markets (equities, FX) and a rolling second test window; (3) adversarial training and multi-source input validation.

14. Research-paper plan

Contribution statement (draft)

"We introduce a leakage-free hourly benchmark for pairwise crypto shock contagion (30 coins, 1.46M events, 2020–2026) and show that temporal graph networks outperform a fitted independent-cascade model at every horizon and outperform gradient boosting out of sample, specifically after market-wide shocks. Faithful TempME explanations reveal that the contagion signal is carried by coins' current state rather than by chains of past events, consistent with same-hour market-wide transmission."

Target venues

- **ACM ICAIF** (International Conference on AI in Finance) — best fit for applied finance + ML.
- **Finance / time-series workshops** at NeurIPS, ICML or KDD — shorter papers, faster review.
- Applied journals later (e.g. finance-and-data-science journals) once a second test window is added.
- Check each venue's current call for papers for deadlines and page limits.

What to add first

1. **Rolling second test window** (e.g. retrain to 2023, test 2024; retrain to 2024, test 2025–26) to show the result is not one period — about 1 day of work plus compute.
2. **Related-work section:** temporal GNNs (TGN, DyGFormer, TGB benchmarks), financial contagion and spillovers (Diebold–Yilmaz, network finance), crypto connectedness studies, explainability (TempME, integrated gradients).
3. Optional: one more market, or sentiment features, to broaden the claim.

Paper outline (8 pages)

1. Introduction: contagion in 24/7 crypto markets; contributions.
2. Related work.
3. Data and event construction (shocks, events, labels, leakage controls).
4. Network analysis findings that motivate the model (market mode, simultaneous transmission).
5. Models and protocol (TGN, DyGFormer, baselines; splits, seeds, bootstrap).
6. Results (main table, significance, by shock type, top-alert precision).
7. Explanations and robustness (TempME diagnostic, integrated gradients, stress tests).
8. Limitations and conclusion.

Timeline: about 4–6 weeks — week 1 rolling window, weeks 2–3 writing and figures, week 4 supervisor review, then submission. Confidentiality: decide on any patent filing **before** submitting, since publication is public disclosure.

15. The professor's questions — spoken answers

How are you quantifying events?

> **Say it (30 s):** A shock is an hourly move bigger than 2.29 times the coin's own recent volatility, computed only from past data, with the threshold chosen on training data. Each shock of one coin becomes an event towards every other coin — 1.46 million events — each carrying four informative features like the pair's correlation and the size of the move. The label is whether the other coin shocks within 1, 3, 6, 12 or 24 hours. We also classify shocks as market-wide, partial or idiosyncratic, and as crashes or pumps.

What models are you using?

> **Say it (30 s):** Our main model is a Temporal Graph Network, and DyGFormer is a second temporal graph model. Against them we compare gradient boosting on the same inputs, an LSTM, the independent cascade diffusion model, and HAR-RV and rule baselines. Diebold–Yilmaz spillovers give us network rankings, TempME and integrated gradients explain the model, and a temporal graph autoencoder looks for anomalous days.

Are you doing neural temporal analysis?

> **Say it (30 s):** Yes, it is the core of the project. The TGN has continuous-time memory for every coin, DyGFormer is a temporal transformer over interaction histories, the LSTM models sequences, the T-GAE uses a GRU over daily graphs, and TempME explains predictions with temporal motifs. Everything is trained in strict time order: memory is only updated after each prediction, so no model ever sees the future.

Are you doing social network analysis?

> **Say it (30 s):** Yes, applied to the market network: coins are the actors and co-movement or shock transmission are the ties. We use correlation networks, minimum spanning trees, Diebold–Yilmaz spillover networks, PageRank and eigenvector centrality, Louvain communities compared with sectors, and the independent cascade model of diffusion. We don't use social-media data — that would be a separate data source.

Is it research-paper worthy?

> **Say it (30 s):** For an applied venue such as ICAIF or a finance workshop, yes: the protocol is rigorous, the graph models significantly beat the cascade and beat plain machine learning out of sample, the top alerts are useful, and the explanations tell us something new about how crypto contagion works. For a top venue we would need a second test window and possibly a second market.

16. When you don't know the answer

Pattern	Example
Scope	"That was outside our scope, but I would expect X because Y." — e.g. order-book data: "We didn't use it, but I'd expect it to help at the 1-hour horizon because shocks are often preceded by liquidity drops."
We tested it	"We tried that and it didn't help: …" — futures data, refit on more recent data, LR schedules, regularisation, T-GAE designs.
Data limit	"Our data can't answer that because …" — e.g. causality of news: we have prices only, no timestamps of news.
Redirect to a result	"What we can say is …" — move to the closest finding you know well (e.g. shock types, top-alert precision).
Name the experiment	"The experiment that would settle it is …" — e.g. a rolling test window, or adding sentiment features and comparing on validation.

Never invent a number. "I don't have that number here, it is in notebook 0X" is a good answer.

17. Self-test — 16 likely questions (cover the answers)

1. Why AUPRC and not accuracy?
2. Why is the shock threshold 2.29?
3. How do you prevent data leakage?
4. Did you look at the test set more than once?
5. Why a temporal graph network rather than an LSTM?
6. What does the independent cascade do, and why is it the baseline?
7. Is the improvement significant?
8. Why generalized FEVD, not Cholesky?
9. What did the ablation show?
10. Does the graph beat plain ML?
11. What did TempME show?
12. How useful are the alerts in practice?
13. Is the model overfitting?
14. How long do shocks last?
15. How robust is the model?
16. What would you do next?

Answers

#	Short answer
1	90% of events are negatives at 1 h; always-no is 90% accurate. AUPRC focuses on positives; random = positive rate (0.105 at 1 h).
2	Chosen on training data only so about 4% of training hours are shocks; relative to each coin's trailing 7-day volatility, so comparable across coins and leakage-free.
3	Fixed-date splits + 7-day embargo; train-only scaling and seasonality; trailing-only features and shocks; strictly-later labels; TGN memory updated only after prediction (checked by integration tests on the real data).
4	Several configurations were each scored once (v1, v2, ablations, refit...). Each was fixed on validation first, v1 → v2 came from validation curves, the main model was never switched for a better test score, and all versions are reported. A rolling second test window is the clean fix.
5	Contagion is about interactions between coins; the LSTM sees one coin. The TGN's memory and event attention model who interacts with whom in continuous time.
6	Classic diffusion model: each shocked coin activates each other coin with a fitted probability. Beating it shows learning beyond counting shocks; TGN 0.205 vs 0.159 at 1 h.
7	Yes: day-block bootstrap, 971 days × 1,000 resamples; e.g. +0.046 [0.019, 0.074] at 1 h vs the cascade.
8	Cholesky depends on the ordering of the 30 coins; generalized FEVD does not.
9	Without memory: same at 1 h, worse at 6–24 h (significant) — memory captures multi-hour contagion.
10	On test yes at 1–3 h ($p \approx 0.03$), and clearly after market-wide shocks; tied after idiosyncratic shocks.
11	Barely above random, because the TGN's attention is uniform and motifs match a shuffled null; signal is in coin state (integrated gradients: receiver 65%).
12	Top 1%: 61% come true within 3 h, 76% within 6 h (random 21% / 31%); 13 of 15 riskiest in the three crashes.
13	Somewhat at 1–3 h (fit ratio 1.5 vs 1.1 for the cascade), controlled by early stopping; none at 24 h.
14	Median 8–10 h to normal volatility; 55–65% re-shock within 24 h.
15	Missing 50% of events: < 0.5% change; noise similar to boosting; vulnerable to white-box FGSM (−35% at $\varepsilon = 0.1$).
16	Rolling second test window, sentiment features, another market, adversarial training.

18. Glossary

Term	Meaning
AUPRC / AUROC	Area under the precision–recall / ROC curve; AUPRC random = positive rate, AUROC random = 0.5
Embargo	Gap between train / validation / test so labels cannot overlap periods
Shock (z)	|deseasonalised hourly return| / trailing 7-day volatility > 2.29
Event	Coin i shocks at hour t → one event i → j for every other coin j
TGN memory	A vector per coin updated by a GRU after every event it takes part in
Focal loss	Loss that down-weights easy examples to focus on rare positives
Day-block bootstrap	Resampling whole days to get honest confidence intervals
Diebold–Yilmaz (GFEVD)	Share of a coin's forecast uncertainty due to shocks in other coins; order-invariant
Market mode	The common factor moving all coins (64% of hourly variance)
Louvain / NMI	Community detection by modularity / similarity of two groupings (0–1)
Independent cascade	Diffusion model: each active node gets one chance to activate each neighbour
TempME	Explainer using temporal motifs and an information-bottleneck objective
Integrated gradients	Attribution summing gradients along a path from a baseline to the input
Characterization score	Harmonic mean of Fidelity+ and 1 − Fidelity−
T-GAE	Temporal graph autoencoder; anomaly = prediction / reconstruction error
FGSM	Fast gradient sign method: worst-case small perturbation of the inputs
Fresh shock	No shock of the same coin in the previous 24 hours